

SIMATS ENGINEERING

ASSESSMENT TOOL 2

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Register Number	192465050
Date	
Time	1 Hr

COURSE NAME: **Cybersecurity Incident Handling And Response**
COURSE CODE: **CSA6001**

***C05:** Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability. (BL3, BL4)*

Activity Tool : Pseudocode Writing Task (Algorithm Design) (Medium)
Assessment Tool Weightage: 20%

Code of Conduct:

*I Abinеш M 192465050 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

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RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Incident Classification and Severity Assessment	25%	Accurately classifies incidents and determines severity based on impact, affected assets, data sensitivity, urgency, and business criticality with clear justification.	Correctly classifies most incidents and determines appropriate severity with minor gaps.	Provides basic classification and severity assessment but misses important factors.	Unable to correctly classify incidents or determine appropriate severity.
Incident Ticket Creation and Documentation	20%	Creates complete and accurate tickets containing incident details, timestamps, affected assets, IOCs, evidence, assigned personnel, actions, and status updates.	Creates well-structured tickets with most required information and minor omissions.	Creates basic tickets but important information or evidence is missing.	Unable to create accurate or complete incident tickets.
Incident Prioritization, Tracking, and Escalation	25%	Effectively prioritizes, tracks, updates, and escalates incidents based on severity, SLA requirements, business impact, and escalation procedures.	Correctly prioritizes and tracks incidents with minor gaps in escalation or SLA handling.	Performs basic tracking and prioritization but misses important escalation or SLA requirements.	Unable to appropriately prioritize, track, or escalate incidents.
SOC Coordination and Incident Resolution	15%	Effectively coordinates with SOC and relevant technical teams, documents response actions, verifies resolution, and properly closes incident tickets.	Coordinates with relevant teams and completes incident resolution with minor omissions.	Demonstrates limited coordination and incomplete resolution or closure procedures.	Unable to coordinate effectively or complete the incident resolution process.
Reporting, Documentation, and Presentation	15%	Provides clear, professional incident reports and ticket records with complete evidence, activity history, SLA status, resolution details, and conclusions.	Documentation is well organized and includes most required information with minor issues.	Documentation is partially organized and lacks important details or evidence.	Documentation is incomplete, poorly organized, and lacks clear incident records or conclusions.

Incident Classification, Ticketing, and Prioritization

Question 1: Incident Classification and Ticket Creation

Receiving a phishing attempt, a malware infection, and unauthorized access to a critical server in the same window requires an incident response analyst to quickly classify each event correctly, since misclassification leads directly to mis-prioritized response effort. My approach is to apply a consistent classification taxonomy to each alert, independently assess severity and priority, and then create a well-structured ticket for each incident containing everything the next responder needs to act without having to re-gather basic context.

1. Classifying Each Incident

- **Phishing attempt:** Classified under the Social Engineering / Initial Access category. Key sub-classification questions: was the email merely received, or did a user click a link or enter credentials? This determines whether it is classified as an attempted or a successful compromise.
- **Malware infection:** Classified under the Malicious Code category, further sub-classified by malware type where known (trojan, ransomware, worm, spyware) since this affects the appropriate response playbook and urgency.
- **Unauthorized access to a critical server:** Classified under the Unauthorized Access / Intrusion category — and, given the asset's criticality, treated as a potential Confirmed Compromise pending verification, since this category carries the highest inherent risk of the three.

Using a standard taxonomy (many organizations align with NIST SP 800-61 categories: Denial of Service, Malicious Code, Unauthorized Access, Inappropriate Usage, Multiple Component) ensures incidents are tracked consistently over time and allows meaningful trend reporting across incident types.

2. Determining Severity and Priority

Severity and priority are related but distinct: severity reflects the technical and business impact of the incident itself, while priority reflects how urgently it demands response resources relative to other current incidents. I calculate both using a consistent formula: $\text{Priority} = \text{Severity (impact)} \times \text{Urgency (rate of potential damage or spread)}$.

Incident	Severity Factors	Urgency Factors	Resulting Priority
Phishing attempt (no click confirmed)	Low — no confirmed compromise yet	Medium — other users may receive it too	Medium
Malware infection (workstation)	Medium — single endpoint, contained	Medium-High — risk of lateral spread	High

Incident	Severity Factors	Urgency Factors	Resulting Priority
Unauthorized access, critical server	High — critical asset, likely sensitive data	High — active access, ongoing risk	Critical

3. Creating the Incident Tickets

Each incident receives its own ticket in the incident tracking system, even where a shared root cause is suspected, since each represents a distinct technical event requiring its own investigation trail. A well-formed ticket includes:

- Unique ticket ID, creation timestamp, and reporting source (SIEM alert, user report, EDR detection).
- Incident category and sub-category (per the classification taxonomy above).
- Severity and priority rating, with the specific factors that drove the rating documented for later audit.
- Affected assets: hostname, IP address, user account, asset owner, and business criticality tier.
- Initial indicators/evidence: relevant log excerpts, alert IDs, file hashes, or email headers already gathered.
- Assigned analyst/team and current status (New, Investigating, Contained, Resolved, Closed).
- A running activity log/comment thread capturing every investigative step and finding as the case progresses.

Where multiple tickets are later confirmed to share a root cause (for example, if the malware infection turns out to have originated from the same phishing email), they are linked as related tickets under a parent incident record rather than merged, preserving each one's individual investigation trail while making the relationship visible to anyone reviewing the case.

4. Example Ticket Template

A representative ticket for the unauthorized-access incident illustrates the level of detail expected at creation:

Field	Example Entry
Ticket ID	INC-2026-0842
Category / Sub-category	Unauthorized Access — Privileged Account Compromise
Severity / Priority	High / Critical (P1)
Affected Asset	SRV-DB-PROD-03 (Tier 1 — production database server)
Detection Source	SIEM correlation alert — privilege escalation + off-hours login
Initial Evidence	Event ID 4672 log excerpt, source IP, session ID attached

Field	Example Entry
Assigned To / Status	Senior Analyst — J. Rivera / Investigating

Capturing this level of structured detail at ticket creation means the next analyst — or an auditor reviewing the case months later — can understand the incident's full context without needing to track down the original reporter or re-query the SIEM.

5. Tools Supporting Classification and Ticketing

- **SIEM / SOAR platforms (Splunk, Sentinel, TheHive, Cortex XSOAR):** generate the initial alert, and increasingly can auto-populate a draft ticket with relevant log context, reducing manual data entry and transcription errors.
- **IT service management (ITSM) tools (ServiceNow, Jira Service Management):** provide the ticketing backbone, workflow states, and SLA tracking, often integrated directly with the SIEM/SOAR platform for seamless handoff.
- **Asset and CMDB integration:** Automatically pulls asset criticality, owner, and business-unit metadata into the ticket so severity assessment does not depend on the analyst's memory of which systems are critical.
- **Threat intelligence enrichment plugins:** Automatically attach reputation data for any IPs, domains, or hashes referenced in the ticket, giving the assigned analyst immediate context without a separate lookup step.

6. Escalation Criteria

Clear, pre-defined escalation triggers ensure a ticket does not sit with a single analyst past the point where it should involve broader response resources:

- Any ticket classified Critical (P1) automatically pages the on-call incident response lead in addition to the assigned analyst.
- A ticket's severity is escalated if investigation reveals broader scope than initially assessed (e.g., a single-host malware ticket found to be part of a multi-host outbreak).
- Any incident involving confirmed regulated-data exposure is escalated to legal and compliance stakeholders regardless of its technical severity rating.
- Tickets approaching SLA breach are automatically flagged to the team lead for reassignment or additional support before the deadline is missed.

Question 2: Incident Tracking and Prioritization

When a security operations team is handling several incidents at once with limited analyst capacity, the incident tracking and ticket management system becomes the central tool for making sure the most damaging incidents get attention first, without losing visibility into the ones that must wait. My approach combines a structured prioritization model with disciplined, continuous ticket updates throughout each incident's lifecycle.

1. Prioritization Factors

- **Severity:** The technical and business impact of the incident if left unaddressed — data breach potential, system outage, or unauthorized privileged access all raise severity.
- **Business impact:** How directly the affected system or process supports revenue, regulatory compliance, or critical operations, since a lower-severity technical issue on a critical system can still outrank a higher-severity issue on a minor one.
- **Affected assets:** The number and criticality tier of hosts, accounts, or services involved — a single low-value workstation ranks differently than a domain controller or production database.
- **Data sensitivity:** Whether the incident involves regulated data (PII, PHI, payment card data, trade secrets), which can elevate priority regardless of technical severity due to legal/regulatory exposure.
- **Urgency:** How quickly the situation is likely to worsen — active data exfiltration or spreading malware demands faster response than a contained, static issue.

2. Using the Tracking System to Prioritize

- **Centralized queue with computed priority scores:** The ticketing system (ServiceNow, Jira Service Management, TheHive, or similar SOAR-integrated platform) calculates a composite priority score from the factors above, so analysts triage from a single ranked queue rather than judging each incident ad hoc.
- **SLA timers tied to priority:** Each priority tier is linked to a defined response and resolution SLA (e.g., Critical: 15-minute acknowledgment, 4-hour containment target), and the system visibly flags tickets approaching or breaching their SLA so at-risk incidents are not silently deprioritized.
- **Dynamic re-prioritization:** As new evidence emerges — for example, a 'medium' malware ticket turning out to involve data exfiltration — the analyst updates the severity/urgency fields, and the system automatically recalculates and re-ranks the ticket's position in the queue.
- **Resource and assignment visibility:** The system shows current analyst workload and skill assignments, allowing a team lead to rebalance active tickets when one analyst is overloaded with high-priority cases while another has capacity.

3. Example Prioritization Matrix

Priority	Example Criteria	Response SLA	Typical Assignment
Critical (P1)	Active breach of critical asset with sensitive data exposure	Immediate (≤15 min)	Senior analyst + IR lead paged
High (P2)	Confirmed malware with lateral-spread risk	≤1 hour	On-shift analyst, escalation path defined
Medium (P3)	Contained incident on non-critical asset	≤8 hours	Standard queue assignment
Low (P4)	Policy violation, low-risk false-positive review	≤3 business days	Batched for routine review

4. Information Recorded and Updated Throughout the Lifecycle

Lifecycle Stage	Information Recorded
Identification	Detection source, initial indicators, reporter, timestamp, preliminary classification
Triage	Severity/urgency scoring, assigned priority, affected asset criticality, assigned analyst
Investigation	Evidence collected, findings, updated scope, revised severity if warranted, related tickets linked
Containment/Eradication	Actions taken, systems isolated, timestamps of each containment step, approvals obtained
Recovery	Restoration steps, validation results, systems returned to production, monitoring period defined
Closure	Final root cause, resolution summary, lessons learned, ticket closure approval and date

Maintaining this level of continuous, structured detail throughout the ticket's lifecycle — rather than only documenting at closure — gives the team lead real-time visibility into true incident status, supports accurate re-prioritization as situations evolve, and produces a complete, audit-ready record once the incident is closed.

5. Managing Limited Resources Across Simultaneous Incidents

- **Triage board / Kanban view:** Most ticketing systems offer a visual board (New → Triaged → Investigating → Contained → Resolved) filtered by priority, letting the team lead see at a glance how many Critical/High tickets are open versus how many analysts are available.
- **Capacity-aware assignment:** Rather than assigning tickets purely by arrival order, the team lead balances assignment against each analyst's current open-ticket load and specialization (e.g., malware analysis vs. network forensics).
- **Temporary re-scoping of lower priority work:** When multiple Critical incidents are active simultaneously, Medium/Low priority tickets are explicitly placed on hold with a documented reason, rather than silently neglected, preserving accountability for eventual follow-up.
- **Shift handoff notes:** For incidents spanning multiple shifts, the ticket's comment log doubles as a structured handoff document, ensuring the incoming analyst has full context without a lengthy verbal briefing.

6. Reporting and Metrics from the Tracking System

Beyond managing individual incidents, the aggregated ticket data supports ongoing security operations reporting that helps justify staffing and tooling investments:

Metric	What It Shows
Mean Time to Acknowledge (MTTA)	How quickly incidents are picked up after ticket creation
Mean Time to Contain (MTTC)	Effectiveness of containment processes across incident types
SLA compliance rate	Percentage of incidents handled within their priority-tier SLA
Incident volume by category	Trends that inform where to invest in additional preventive controls

Reviewing these metrics regularly also surfaces systemic bottlenecks — for example, consistently missed SLAs on a specific incident category may indicate a need for additional training, tooling, or headcount in that area, turning routine ticket data into a driver of continuous program improvement.

Conclusion

Both scenarios rely on the same foundation: a consistent classification and prioritization framework, applied through a disciplined ticketing system, is what allows a security team to handle multiple simultaneous incidents rationally rather than reactively. Correctly classifying and scoring each incident at intake ensures response effort is allocated where it matters most, while continuous, structured updates throughout the ticket lifecycle keep that prioritization accurate as new information emerges. Together, these practices turn a chaotic multi-incident situation into a manageable, auditable

workflow — ensuring that limited analyst resources are always pointed at the incidents posing the greatest risk to the organization.

Comparing the Two Scenarios

Aspect	Question 1: Classification & Ticketing	Question 2: Tracking & Prioritization
Primary focus	Correctly categorizing and documenting each new incident	Ranking and managing many open incidents against limited resources
Key output	A complete, well-structured ticket per incident	A dynamically ranked queue with SLA tracking
Main risk if done poorly	Misclassified incidents routed to the wrong response process	Critical incidents delayed behind lower-impact work
Supporting data	Evidence, affected assets, initial severity rating	Business impact, data sensitivity, real-time status updates

In practice, these two capabilities operate continuously together: every new alert must first be correctly classified and ticketed before it can be meaningfully prioritized against the team's existing workload, and the tracking system's ongoing prioritization depends entirely on the quality and completeness of the information captured at ticket creation. Investing in both — a rigorous intake process and a disciplined, metrics-driven tracking system — is what allows a resource-constrained security team to consistently respond to what matters most, first.